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Getting started

Everything you need for this course runs inside a Docker container: Python, JupyterLab, scikit-learn, XGBoost and TensorFlow are already installed. You do not install Python on your machine.

You will never need an API key or a paid account for this course.

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1. Install Docker Desktop

Download it from docker.com and start it. Windows, macOS and Linux are all fine.

You need roughly **20 GB of free disk space** and **8 GB of RAM**.

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2. Clone the course repository

```
git clone https://github.com/fabioantonini/technologies-for-artificial-intelligence.git
cd technologies-for-artificial-intelligence
```

This is how you will receive new lessons throughout the course, so clone it somewhere you will find again — not your Downloads folder.

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3. Start the environment

From inside the repository folder:

```
docker compose up
```

The first run downloads the image and takes several minutes. Later runs start in seconds.

Then open:

```
http://127.0.0.1:8888/lab?token=aicourse
```

JupyterLab opens with the course material already there. Anything you save is written to your own machine, inside the folder you cloned.

To stop it, press `Ctrl+C` in the terminal, or run `docker compose down`.

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4. Getting each new lesson

New material is published on the day of each lesson. To fetch it:

```
git pull
```

That is all. A few megabytes, a couple of seconds — **you do not re-download the Docker image.** The image only changes when the software environment does, which is rare, and you will be told when it happens.

> **If you have edited a course notebook** and `git pull` complains about a conflict, the simplest fix is to copy your version to a new name (`my_lesson3.ipynb`), then run `git checkout -- .` and pull again. Better still: work on copies from the start, and keep your own work in a `my_work/` folder.

Troubleshooting

Port 8888 is already in use. Something else is using it. Either stop that, or edit `docker-compose.yml` and change `"8888:8888"` to `"8889:8888"`, then open `http://127.0.0.1:8889/lab?token=a`

"Permission denied" when saving a notebook. This should not happen — the compose file is configured to prevent it. If it does, stop the container, run `docker compose down`, then `docker compose up` again. Report it if it persists.

The container is slow to start the first time. Expected: the image is initialising and the environment is large. Subsequent starts are much faster.

I deleted my container, is my work gone? No. Your files live in the folder you cloned, on your own machine. Containers are disposable; your work is not.

Do I need a GPU? No. Every lab in this course runs on CPU by design.

Running without cloning (not recommended)

The published image contains a snapshot of the course, so this works:

```
docker run --rm -p 8888:8888 -e JUPYTER_TOKEN=aicourse \
  fabioantonini/technologies-for-artificial-intelligence:0.1.0
```

But your work is stored **inside the container and lost when it stops**, and you have to pull a multi-gigabyte image for every new lesson. Use the `git clone` route above.